Graphical Abstract

V-Sparse: From Temporal-Spatial Visual Semantic Compression to Coarse-to-Fine Interaction for Text-Video Retrieval

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Highlights

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- A novel visual compression and interaction framework for text-video retrieval.
- A text-guided dynamic visual compression module for representation enhancement.
- A novel coarse-to-fine interaction module for integrating diverse modal alignments.
- Experiments on six benchmarks demonstrate that V-Sparse achieves SOTA performance.

V-Sparse: From Temporal-Spatial Visual Semantic Compression to Coarse-to-Fine Interaction for Text-Video Retrieval

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Abstract

Text-to-video retrieval refers to the task of finding the most relevant videos in a large-scale unlabeled video collection based on a given text query. In recent years, CLIP-based text-video retrieval methods have developed rapidly, with research primarily focusing on feature-enhancement techniques and interaction strategies. However, due to the concise nature of text and the rich modalities of video, computing similarity scores alone is insufficient for high-precision cross-modal retrieval. To address the inherent imbalance in cross-modal matching, we propose a novel text-video retrieval model, named **V-Sparse**, which includes visual semantic compression for feature enhancement and coarse-to-fine alignment for feature interaction. First, we propose a text-guided **V**isual **S**emantic **C**ompression (**VSC**) module, consisting of **T**emporal (**TVSC**) frame-level and **S**patial (**SVSC**) patch-level compres-

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sion, aimed at reducing feature redundancy and providing precision support for coarse-to-fine interaction. Second, benefiting from visual semantic compression, we propose a novel **C**oarse-to-**F**ine granularity **I**nteraction module (**CFI**), which aligns sentences with frames, sentences with patches, and words with patches from a unified joint feature encoding perspective. VSC and CFI jointly facilitate cross-modal text-video alignment from the perspectives of feature enhancement and feature interaction, greatly mitigating the inherent imbalance in modal pairing. We evaluate the performance of V-Sparse on six benchmark datasets and achieve state-of-the-art results in both long-video and short-text retrieval. Importantly, V-Sparse demonstrates the importance of feature compression in cross-modal interaction through extensive ablations and offers an effective intermediate pathway for modality interaction. Code will be available at https://github.com/OPA067/V-Sparse.

Keywords: Cross-modal representation learning, text-to-video retrieval, video semantic compression, feature granularity interaction.

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1. Introduction

With the rapid development of the Internet, especially video platforms such as YouTube, massive amounts of unlabeled video data are continuously uploaded and shared. Text-video retrieval, the task of finding the most relevant video based on a given text query, has gained increasing research attention due to its practical value in facilitating video content dissemination. Recently, the large-scale vision-language pre-training model CLIP [1] has achieved significant success in numerous multimodal tasks, *e.g.*, visual question answering [2], classification [3], and image retrieval [4], demonstrating its strong vision-language alignment capabilities. Existing video retrieval methods [5, 6] directly use CLIP to project text and video into a common space, establishing feature-level similarity relationships. For example, Clip4Clip [7] transfers the image-language knowledge of the CLIP model to video-language retrieval in an end-to-end manner, establishing a baseline for video retrieval.

Currently, text-video retrieval methods mainly focus on two aspects: feature enhancement [8, 9] and granularity interaction [10, 11]. Feature enhancement aims to strengthen visual or textual features to improve cross-modal matching accuracy. For example, X-Pool [12] employs an attention mechanism to extract the most relevant video frames for a given text, thereby enhancing the representational capacity of visual feature. Granularity interaction focuses on aligning information across coarse and fine levels. UCoFiA [13] proposes a unified text-visual coarse-to-fine alignment model to improve text-video retrieval, while HBI [14] builds hierarchical frame-level and word-level representations for fine-grained alignment, attracting considerable attention. However, these methods overlook the inherent differences between text and video modalities and do not account for the accuracy of feature interaction at both granularities, which can lead to inaccurate retrieval results.

In Figure 1, we illustrate the above failure scenario using a specific query example. Given the text query0771 *"a little girl does gymnastics"* in the MSRVTT datasets [15], Figure 1(a) shows the failed retrieval result using the frame-based feature enhancement model X-Pool [12], where the correct query is *"video of gymnasts practicing to roll"*. Figure 1(b) displays another failed retrieval result using the fine-grained word-patch interaction model Clip4clip [7], where the correct query is *"some girls are practicing gymnastics"*. Nei-

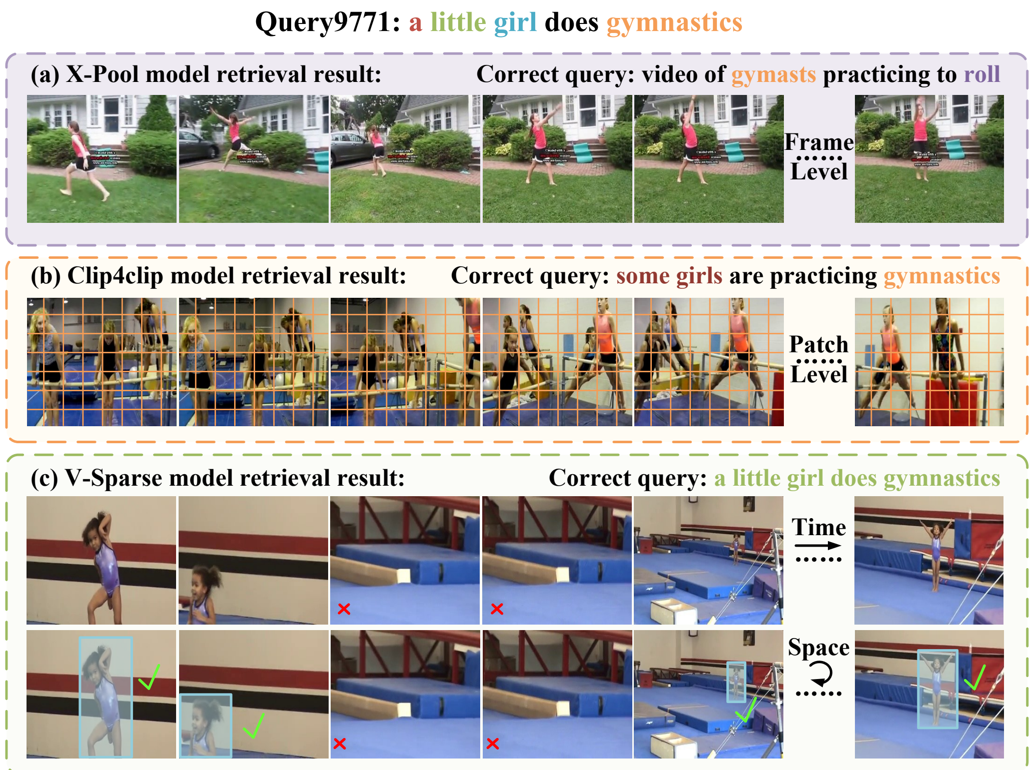


Figure 1: **Motivation.** Taking query9771 as an example, we illustrate the limitations of existing model under sparse text and redundant visual features. (a, b) Failed retrieval results from the frame-based feature enhancement model X-Pool [12] and the patch-based feature interaction model Clip4Clip [7]. (c) Successful retrieval results from our model V-Sparse, which benefits from text-supported temporal-spatial visual semantic compression and adaptive coarse-to-fine feature interaction.

ther feature enhancement nor granularity interaction is able to retrieve the target video, as shown in Figure 1(c). The failure occurs because the target video contains invalid frames in the temporal dimension (*e.g., "a little girl is missing from the frame"*) and irrelevant regions in the spatial dimension (*e.g., "numerous training scenes"*). This represents one of the most challenging problems in video retrieval, as allowing all visual content to interact with text directly makes it difficult to maintain consistent feature accuracy.

The above analysis indicates that effective text-video retrieval requires selectively processing visual features rather than treating all of them equally. Therefore, at the bottom of Figure 1, we manually identify the visual entities required for interaction with the text (✓) and the visual content to be excluded (✗). Additionally, after obtaining the compressed visual features, further interaction with the text is required, considering both global alignment at the macro level and local alignment at the micro level, to avoid the single-interaction pitfalls illustrated in Figure 1(a) and (b).

Motivated by the discussion above, we introduce **V-Sparse**, a novel text-video retrieval model that progresses from temporal-spatial visual compression to coarse-to-fine interaction. Figures 2 and 4 illustrate the video semantic compression and coarse-to-fine alignment in the V-Sparse model, respectively. **First**, we leverage CLIP's powerful semantic feature extraction capabilities to obtain text-level sentence and word features, as well as video-level frame and patch features, for subsequent feature enhancement and interaction. **Second**, we propose a text-guided **V**isual **S**emantic **C**ompression (**VSC**) module, consisting of **T**emporal (**TVSC**) frame-level and **S**patial (**SVSC**) patch-level compression, aiming to reduce feature redundancy and providing precision support for coarse-to-fine interaction. For TVSC, we dynamically select the video frames most relevant to the text query based on global sentence-frame similarity, effectively eliminating redundant frames and patches. For SVSC, we use a K-Nearest-Neighbor-based density peaks clustering algorithm (DPC-KNN) [16], guided by sentence features, to dynamically aggregate patch features within spatial regions. Both feature compression strategies are guided by text, effectively constructing visual features that are highly aligned with the text query. **Third**, benefiting from visual semantic compression, we propose a novel **C**oarse-to-**F**ine granularity **I**nteraction module (**CFI**), which aligns sentences with frames, sentences with patches, and words with patches from a unified joint feature encoding perspective. These interaction strategies jointly enhance text-video retrieval performance by optimizing feature compatibility and reducing feature redundancy. The main contributions of this work are summarized as follows:

- We propose a novel text-video retrieval model, named V-Sparse, which includes visual semantic compression for feature enhancement and coarse-to-fine alignment for feature interaction.

- We propose a text-supported video semantic compression module that dynamically compresses visual features, including temporal frames and spatial patches, enhancing the representational power of visual features.

- Benefiting from visual semantic compression, we introduce a novel coarse-to-fine interaction module, aligning sentences with frames and patches, and words with patches in an adaptive granularity manner.

- We conduct extensive experiments on six benchmark datasets, including MSRVTT (R@1, 50.9%), DiDeMo, LSMDC, ActivityNet, Charades, and VATEX, and achieve state-of-the-art retrieval results.

2. Related Work

2.1. Text-Video Retrieval

Text-video retrieval aims to retrieve the most semantically relevant video from a large collection based on a given textual query. Early works [10, 17, 5] focus on designing feature enhancement mechanisms to align text and video features while also establishing many benchmarks [15, 18, 19]. Recently, transformer-based text-video retrieval methods [6, 12, 20, 21, 22] utilize cross-attention to abstract multi-modal cues, leading to significant performance gains. For example, Liu *et al.* [6] propose a "token shift and selection transformer" that uses a token shift module to preserve token integrity and capture subtle actions, thereby improving retrieval performance. To learn more discriminative video representations, Gorti *et al.* [12] use an attention mechanism to extract the most relevant video frames corresponding to a given text, improving the matching of similar text-video pairs. Subsequently, Lei *et al.* [5] adopt sparse sampling on video data to enable end-to-end training and utilize text-image pretraining for text-video retrieval. Wu *et al.* [8] employ captioners to generate positive sample texts, effectively addressing data scarcity in text datasets. Additionally, various post-processing techniques [23, 9, 24] have been employed to enhance retrieval performance. Bogolin *et al.* [23] improve cross-modal retrieval stability and accuracy by re-normalizing query similarity scores to address centrality issues.

2.2. Cross-Modal Feature Enhancement

Cross-modal feature enhancement includes both **vision-based** [12, 6] and **text-based** [8, 9] approaches, serving as an important means to strengthen semantic consistency and complementarity across modalities. Specifically, **1)** vision-based feature enhancement focuses on visual representations that are more closely aligned with textual semantics. Chen *et al.* [11] adopts frame adaptation and text-guided attention mechanisms to jointly enhance the effectiveness of multilevel semantic visual features, leading to superior performance. Wang *et al.* [13] effectively represents visual features at both coarse and fine levels and further proposes a unified coarse-to-fine retrieval model that significantly improves retrieval performance. **2)** Text-based feature enhancement aims to expand originally sparse text queries, either implicitly or explicitly. At the implicit level, Wang *et al.* [9] enhances textual feature representation at the implicit semantic level through random text modeling and regularization techniques. At the explicit level, Xue *et al.* [24] performs post-training on CLIP using a large-scale video-text dataset and enriches each sample pair's semantics by incorporating textual information from multiple language sources. In contrast to methods that solely enhance video and text feature representations, V-Sparse performs video feature compression, offering greater flexibility in aligning semantically sparse text. Moreover, it avoids complex text augmentation while improving both granularity alignment accuracy and efficiency in training and sampling.

2.3. Cross-Modal Granularity Interaction

Current cross-modal text-video granularity interaction strategies can be broadly divided into three types: **1)** coarse-grained interaction, **2)** fine-grained interaction, and **3)** coarse-to-fine interaction. For coarse-grained interaction, Jin *et al.* [25] adopt a cross encoding approach to obtain coarse-grained visual features for interaction with text. For fine-grained interaction, Wang *et al.* [26] propose a high-dimensional sparse hashing framework that leverages low-level multimodal features to achieve fine-grained similarity. To capture the semantic relationships between text and video features at both high and low levels, many methods [13, 11, 22, 14] combine coarse- and fine-grained features. Chen *et al.* [11] introduce a multi-level semantic interaction model that uses adaptive frame section and text-guided attention to reduce video redundancy and enhance modality interactions. Wang *et al.* [13] design a unified model with multi-level alignment and interactive similarity modules, significantly enhancing retrieval accuracy. Compared with the above interaction strategies, V-Sparse primarily focuses on the comprehensiveness and precision of feature interaction under visual semantic compression, ensuring that the video features involved in text interaction remain sufficiently well-aligned. Meanwhile, we aim to explore a unified and comprehensive feature interaction mechanism that remains effective regardless of the number of features involved on either side.

3. Methodology

In this section, we introduce the proposed V-Sparse text-video retrieval model step by step. Specifically, Section 3.1 presents some preliminaries, including feature extraction and feature interaction. Section 3.2 presents video semantic compression guided by textual semantics. Section 3.3 describes the coarse-to-fine granularity alignment process. Section 3.4 explains the training objective and inference strategy. Figures 2 and 4 show the model framework.

3.1. Preliminaries

3.1.1. Feature Extraction

Recent works [7, 12, 14, 9] have demonstrated the strong capability of CLIP [1] in multi-modal feature representation, which inspires us to adopt CLIP as our backbone. Given a text query $T \in \mathcal{T}$, we leverage the CLIP text encoder to extract the word features $T_w = [w_{BOS}, w_1, w_2, \ldots, w_{N_w}, w_{EOS}] \in \mathbb{R}^{(N_w+2)\times D}$, where N_w denotes the length of the word sequence, and set the w_{BOS} and w_{EOS} token as the beginning and end of the sequence, respectively. We take the representation of the w_{EOS} token as the sentence feature $T_s = [s] \in \mathbb{R}^{1\times D}$. Similar to the CLIP text encoder, we use the CLIP image encoder to extract video features. Specifically, given a video $V \in \mathcal{V}$ comprising multiple frames, we uniformly sample N_f frames $[f_1, f_2, \ldots, f_{N_f}] \in \mathbb{R}^{N_f\times H\times W\times C}$, where each frame is divided into N_p patches $[p_1, p_2, \ldots, p_{N_p}] \in \mathbb{R}^{N_p\times P\times P\times C}$ with $P \times P$ size. We utilize the CLIP image encoder to extract the patch features $V_p = [p_1, p_2, \ldots, p_{N_p}] \in \mathbb{R}^{N_p\times D}$ for each frame, and set p_0 as the [CLS] token of the current frame. We aggregate the [CLS] tokens of all video frames to obtain the frame features $V_f = [f_1, f_2, \ldots, f_{N_f}] \in \mathbb{R}^{N_f\times D}$. In summary, by leveraging the pretrained feature extractor, we obtain video frame features V_f and patch features V_p, as well as sentence features T_s and word features T_w. The feature extraction process is illustrated in Figure 2 (a).

3.1.2. Feature Interaction

Feature interaction refers to the process of constructing similarity relationships between cross-modal text and video features. We take sentence feature $T_s \in \mathbb{R}^{1\times D}$ and frame features $V_f \in \mathbb{R}^{N_f\times D}$ as an example to illustrate this process. Given T_s and V_f, we directly compute their cosine similarity

S_{T_s,V_f} following Equation 1:

$$S_{T_s,V_f} = \frac{T_s \cdot V_f}{\|T_s\|_2 \cdot \|V_f\|_2}, \quad V_f = \frac{1}{N_f}\sum_{i=1}^{N_f} V_{f,i}, \tag{1}$$

where $V_{f,i}$ denotes the i^{th} feature of V_f. We use a symmetric cross-entropy loss function to calculate the interaction similarity loss $\mathcal{L}_{T_s,V_f}$, and this process is divided into text-to-video and video-to-text two directions:

$$\mathcal{L}_{T_s,V_f} = -\frac{1}{2B}\Bigg(\underbrace{\sum_{i=1}^{B}\log\frac{\exp(S_{T_s^i,V_f^i}/\tau)}{\sum_{j=1}^{B}\exp(S_{T_s^i,V_f^j}/\tau)}}_{\text{sentence-to-frame}} + \underbrace{\sum_{i=1}^{B}\log\frac{\exp(S_{T_s^i,V_f^i}/\tau)}{\sum_{j=1}^{B}\exp(S_{T_s^j,V_f^i}/\tau)}}_{\text{frame-to-sentence}}\Bigg), \tag{2}$$

where B and τ represent the batch size and temperature hyperparameter, respectively, and $S_{T_s^i,V_f^j}$ represents the similarity between the i^{th} text and the j^{th} video. The objective of this loss function is to maximize the similarity for relevant text–video pairs ($i = j$) and to minimize the similarity for irrelevant pairs ($i \neq j$). Similarly, the feature interaction losses across different coarse-to-fine granularities, $\mathcal{L}_{T_s,V_p}$, $\mathcal{L}_{T_w,V_f}$, and $\mathcal{L}_{T_w,V_p}$, can be computed.

3.2. Proposed Method: Visual Semantic Compression

3.2.1. Motivation

In Section 3.1, we can directly use the CLIP feature extractor to perform interaction on the extracted coarse-to-fine granularity features. Existing feature interaction methods include X-Pool [12], which uses a Transformer to achieve coarse-grained interaction between sentences and video frames; HBI [14], which implements hierarchical multi-granularity interaction; and UCoFiA [13], which unifies coarse-to-fine granularity interactions. Although

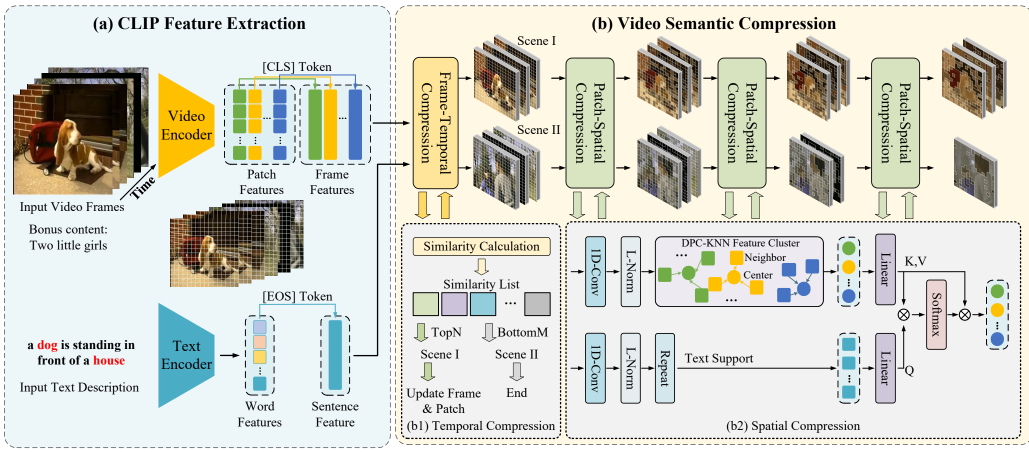


Figure 2: **Framework.** (a) The CLIP Feature Extraction module leverages CLIP [1] to extract both text features (sentence and words) and video features (frames and patches). (b) The Video Semantic Compression module includes (b1) temporal-level frame compression and (b2) spatial-level patch compression. $\otimes$ denote matrix multiplication.

these methods achieve excellent retrieval performance through feature interaction, they do not sufficiently address the *granularity adaptation* between the interacting modalities. Compared to sparse text representations, the visual features V_f and V_p exhibit high redundancy and misalignment across both temporal and spatial dimensions. Therefore, in this section, we propose to perform semantic compression on the visual features from both temporal and spatial dimensions, guided by the query text, to enhance their representational capacity in subsequent granularity interactions.

3.2.2. Temporal Visual Semantic Compression

Intuitively, a video consists of a sequence of frames, each of which is composed of multiple patches. By effectively selecting frames along the temporal distribution, we can not only reduce the number of video frames involved in coarse-grained features but also significantly reduce the number of patches. Based on this, we propose a novel **T**emporal **V**isual **S**emantic **C**ompression module (**TVSC**), which leverages sentence semantics to selectively retain or discard video frame features. Specifically, given the temporal video features $V_f = [f_1, f_2, \ldots, f_{N_f}] \in \mathbb{R}^{N_f \times D}$ and the query sentence feature $T_s = [s] \in \mathbb{R}^{1\times D}$, we can determine which frame features to retain based on their similarity $S_{T_s,V_f} \in \mathbb{R}^{1\times N_f}$ and the number of frames to retain N_f^*:

$$V_f^* = \{f_i \mid i \in \arg\max_i S(T_s, V_f),\ i = 1, 2, \ldots, N_f^*\}. \tag{3}$$

In addition, the corresponding patch features can be updated as:

$$V_p^* = \{p_j \mid j \in \mathcal{I}(V_f^*)\}, \tag{4}$$

where $\mathcal{I}(V_f^*)$ denotes the set of patch indices corresponding to $V_f^* \in \mathbb{R}^{N_f^* \times D}$.

3.2.3. Spatial Visual Semantic Compression

Temporal compression along the frame dimension can greatly reduce the number of frame and patch features involved in subsequent feature interactions. For example, by setting $N_f = 12$ and $N_f^* = 6$, the number of patch features before and after temporal compression is $N_f \times N_p = 588$ and $N_f^* \times N_p = 294$, where $N_p = \frac{H\times W\times C}{P\times P\times C} = \frac{224\times224\times3}{32\times32\times3} = 49$. However, directly interacting with these patch features individually introduces significant redundancy. On the one hand, a single patch lacks complete entity information; on the other hand, many patches are irrelevant to the text description. Therefore, it is necessary to merge and compress patch features from the visual spatial dimension to achieve precise alignment with the text description.

To address the spatial redundancy of the aforementioned patch features, we propose a novel **S**patial **V**isual **S**emantic **C**ompression module (**SVSC**), which combines feature clustering and text-supported cross-attention to perform multi-level compression on patch features. Specifically, we utilize a variant of the k-nearest neighbor-based density peaks clustering algorithm (DPC-KNN) [16], to group similar patch features. Given patch features V_p^*, we compute the local density ρ_i of each patch p_i according to its k-neatest neighbors:

$$\rho_i = \exp(-\frac{1}{k} \sum_{p_j \in \mathrm{KNN}(p_i)} ||p_i - p_j||_2), \tag{5}$$

where $\mathrm{KNN}(p_i)$ denotes the k-nearest neighbors of the patch p_i. Then, we compute the distance indicator δ_i of each patch:

$$\delta_i = \begin{cases} \min ||p_i - p_j||_2, & \rho_i < \rho_j, \\ \max ||p_i - p_j||_2, & \rho_i \geq \rho_j. \end{cases} \tag{6}$$

Intuitively, the patch p_i with a larger local density ρ_i and distance indicator δ_i is more likely to become a cluster center. We determine a cluster center by selecting the patches with the highest scores $\rho_i \times \delta_i$, and then merge the neighboring patches. In addition to reducing the number of patch features, we incorporate textual semantic support to enhance the representational quality of the final visual features. The text query T_s is fed to a transformer block as the query Q, while the merged patch features V_p^* are used as the key K and value V, and the attention is computed as follows:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{7}$$

where d_k represents the feature dimension. Through multiple iterations of the compression, the number of patch features can be gradually reduced.

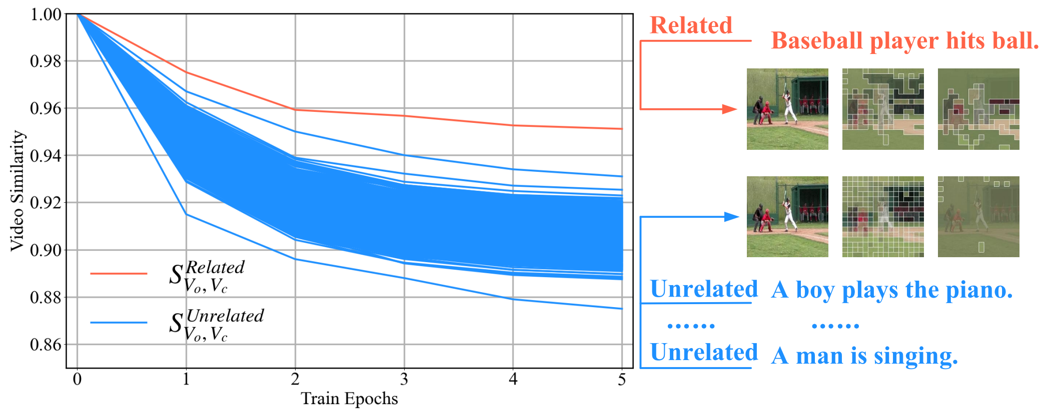


Figure 3: **Similarity**. Video similarity before and after compression. We select video7021 as an example to compare the video similarity before and after compression under 1000 different text conditions. The similarity for the related text-video pair $S^{Related}_{V_o,V_c}$ is marked in red, while that for the 999 unrelated text-video pairs $S^{Unrelated}_{V_o,V_c}$ is marked in blue.

To demonstrate the effectiveness of text-supported compression, a direct approach is to compare the differences in the video features before V_o and after V_c compression under different text conditions. We select 999 texts and one text–video pair from the MSRVTT [15], and plot the similarity relationship between $S^{Related}_{V_o,V_c}$ and $S^{Unrelated}_{V_o,V_c}$ under 1000 text conditions in Figure 3. It can be observed that for related text-video pair, the similarity $S^{Related}_{V_o,V_c}$ is significantly higher than that of unrelated text-video pairs $S^{Unrelated}_{V_o,V_c}$, demonstrating that different texts lead to distinct visual feature representations. As the number of training epochs increases, both types of similarity gradually decrease, indicating a larger semantic gap between V_o and V_c. Additionally, we compare the similarity between the text and video features before and after compression in Section 4.4, which further demonstrates the effectiveness of the compressed visual features.

In summary, the core of video semantic compression lies in enhancing the representations of frame features along the temporal dimension and patch features along the spatial dimension. This process consistently relies on text-conditioned guidance at both global and local levels, exhibiting distinct feature representations under different textual conditions. Although the proposed feature enhancement method for compression can bridge the gap between sparse textual descriptions and rich video content, it has not yet considered how to effectively model the interactions among the compressed features. For example, as shown in Figure 3, how the textual query *"Baseball"* is associated with multiple baseball players in the image is one of the key issues that need to be considered in coarse-to-fine feature alignment.

3.3. Proposed Method: Coarse-to-Fine Feature Interaction

Feature compression and feature alignment are two effective strategies for feature enhancement and interaction. Benefiting from the discriminative representations generated by the feature compression module, we propose a novel **C**oarse-to-**F**ine **I**teraction (**CFI**) module to achieve granularity-adaptive feature interaction in this section. As illustrated in Figure 4, the CFA module includes interactions between coarse-grained sentence and compressed video frame features ($T_s \leftrightarrow V_f^*$), interactions between medium-grained sentence and compressed video patch features ($T_s \leftrightarrow V_p^*$), and interactions between fine-grained words and patch features ($T_w \leftrightarrow V_p^*$). Each type of interaction handles different objects and employs distinct mechanisms.

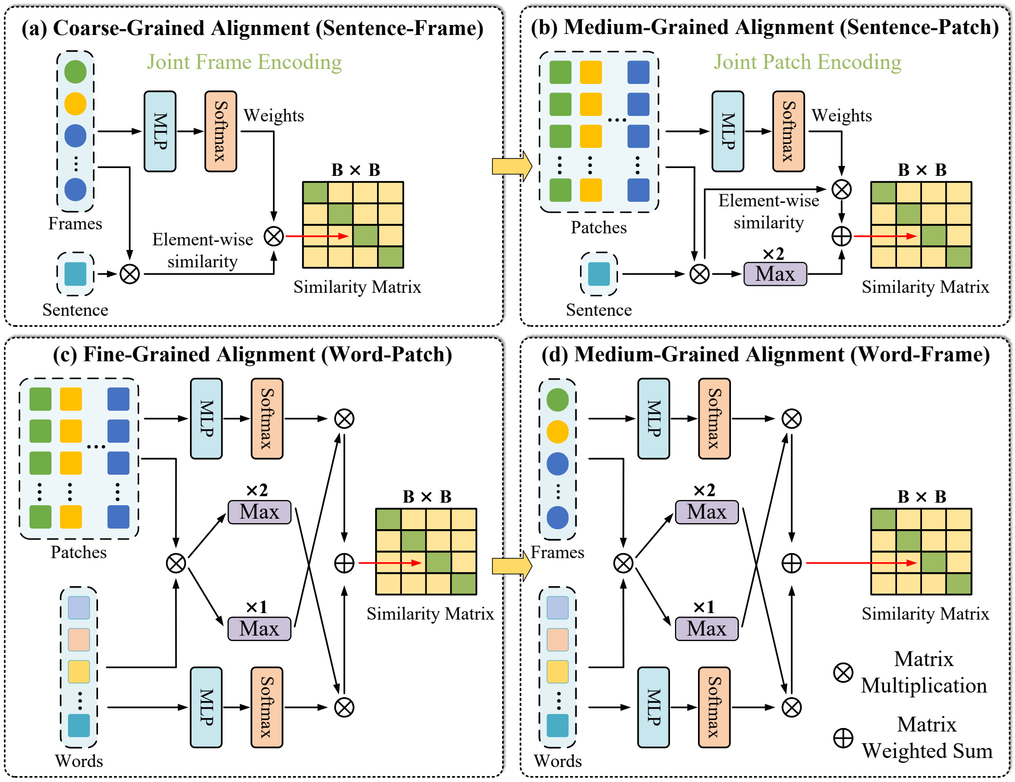


Figure 4: **Framework**. (a) Coarse-grained interaction between a text sentence and a video frame features. (b) Medium-grained interaction between text sentence and video patch features. (c) Fine-grained interaction between sentence words and video patch features. (d) Medium-grained interaction between sentence words and video frame features. $\oplus$ and $\otimes$ denote matrix addition and multiplication, respectively.

3.3.1. Coarse-Grained Interaction

The coarse-grained interaction module is illustrated in Figure 4 (a). Given the interaction objects, a text sentence $T_s = [s] \in \mathbb{R}^{1\times D}$ and video frames $V_f^* = [f_1, f_2, \dots, f_{N_f^*}] \in \mathbb{R}^{N_f^*\times D}$, we employ a joint frame encoding method to compute the similarity S_{T_s,V_f^*} between them:

$$S_{T_s,V_f^*} = \sum_{i=1}^{N_f^*} \theta_f^i a_i, \quad a_i = \frac{s \cdot f_i}{\|s\|_2 \cdot \|f_i\|_2}. \tag{8}$$

where $\theta_f = [\theta_f^1, \theta_f^2, ..., \theta_f^{N_f^*}] = \mathrm{Softmax}(\mathrm{MLP}_f(V_f^*))$ denotes the frame feature weights, and a_i denotes the similarity between s and the i^{th} frame. Unlike the average pooling method used in Equation 1, Equation 8 employs learnable feature weights to identify important video frames, rather than treating all frames equally. Finally, we compute the coarse-grained interaction loss $\mathcal{L}_{T_s,V_f^*}$ according to Equation 2:

$$\mathcal{L}_{T_s,V_f^*} = -\frac{1}{2B}\Bigg(\underbrace{\sum_{i=1}^{B} \log \frac{\exp(S_{T_s^i,V_f^{i*}}/\tau)}{\sum_{j=1}^{B}\exp(S_{T_s^i,V_f^{j*}}/\tau)}}_{\text{sentence-to-frame}} + \underbrace{\sum_{i=1}^{B} \log \frac{\exp(S_{T_s^i,V_f^{i*}}/\tau)}{\sum_{j=1}^{B}\exp(S_{T_s^j,V_f^{i*}}/\tau)}}_{\text{frame-to-sentence}} \Bigg). \tag{9}$$

3.3.2. Medium-Grained Interaction

The medium-grained interaction module is illustrated in Figure 4 (b). Given the interaction objects, a text sentence $T_s = [s] \in \mathbb{R}^{1\times D}$ and video patches $V_p^* = N_f^* \times [p_1, p_2, \ldots, p_{N_p^*}] \in \mathbb{R}^{N_f^* \times N_p^* \times D}$, we extend the coarse-grained interaction to accommodate the newly added patch features:

$$S_{T_s,V_p^*} = \frac{1}{2}\Bigg(\underbrace{\max_i \max_j a_{i,j}}_{\text{sentence-to-patch}} + \underbrace{\sum_{i=1}^{N_f^*}\sum_{j=1}^{N_p^*} \theta_{f,p}^{i,j} a_{i,j}}_{\text{patch-to-sentence}} \Bigg), \tag{10}$$

where $a_{i,j} = \frac{s \cdot p_{f_i,j}}{\|s\|_2 \cdot \|p_{f_i,j}\|_2}$ denotes the similarity between the sentence feature s and the j^{th} patch feature of the i^{th} video frame, $\max\limits_i$ and $\max\limits_j$ denote taking the maximum along the N_f^* and N_p^* dimensions, respectively, and

$\theta_{f,p}^{i,j} = \mathrm{Softmax}(\mathrm{MLP}_{f,p}(V_p^*)) \in \mathbb{R}^{N_f^* \times N_p^*}$ denotes the visual feature weights. Finally, we can compute the medium-grained interaction loss $\mathcal{L}_{T_s,V_p^*}$ according to Equation 2 and Equation 9.

3.3.3. Fine-Grained Interaction

Equations 8 and 10 employ a joint feature encoding approach to enable interactions between the compressed visual features and the global-level text features. However, this process still does not fully leverage word-level features under sparse textual conditions. Since word features are sparsely distributed due to the sparsity of the text content, only a few key words are typically able to effectively describe specific entities in the video, such as people or animals. Therefore, establishing alignment relationships between fine-grained entities is also crucial for accurately capturing the semantic details of the video.

Due to the inherent sparsity of text words and visual patch features, existing retrieval methods rarely consider fine-grained alignment, or only do so in a simplistic manner. For example, although Clip4clip [7] employs an average pooling approach, it overlooks the precision of alignment; while DiCoSA [25] introduces the idea of feature disentanglement, it fails to fully ensure the effectiveness of feature representations. Therefore, we reconsider fine-grained alignment in the context of visual semantic compression. The fine-grained interaction module is illustrated in Figure 4 (c).

Given the interaction objects, text words $T_w = [w_1, w_2, \ldots, w_{N_w}] \in \mathbb{R}^{N_w \times D}$ and video patches $V_p^* = N_f^* \times [p_1, p_2, \ldots, p_{N_p^*}] \in \mathbb{R}^{N_f^* \times N_p^* \times D}$, We adopt the joint encoding approach from the coarse- and medium-grained interactions to compute the similarity S_{T_w,V_p^*} between them:

$$S_{T_w,V_p^*} = \frac{1}{2}\Bigg(\underbrace{\sum_{i=1}^{N_w} \theta_w^i \max_j \max_k a_{i,j,k}}_{\text{word-to-patch}} + \underbrace{\sum_{j=1}^{N_f^*} \sum_{k=1}^{N_p^*} \theta_{f,p}^{i,j} \max_i a_{i,j,k}}_{\text{patch-to-word}} \Bigg), \tag{11}$$

where $a_{i,j,k}$ denotes the similarity between the i^{th} word feature in T_w and the k^{th} patch feature of the j^{th} video frame in V_p^*, and $\theta_w = [\theta_w^1, \theta_w^2, \ldots, \theta_w^{N_w}] = \text{Softmax}(\text{MLP}_w(T_w))$ denotes the word feature weights. Moreover, the method for constructing the similarity between the fine-grained features T_w and V_p^* can also be applied to T_w and V_f^*:

$$S_{T_w,V_f^*} = \frac{1}{2}\Bigg(\underbrace{\sum_{i=1}^{N_w} \theta_w^i \max_j a_{i,j}}_{\text{word-to-frame}} + \underbrace{\sum_{j=1}^{N_f^*} \theta_f^j \max_i a_{i,j}}_{\text{frame-to-word}} \Bigg). \tag{12}$$

Finally, the interaction similarity losses $\mathcal{L}_{T_w,V_p^*}$ and $\mathcal{L}_{T_w,V_f^*}$ can be computed according to Equations 11 and 12, respectively.

3.4. Training and Inference

3.4.1. Training Objects

By integrating the similarities obtained from the coarse-to-fine feature interactions in Section 3.3, we can sequentially obtain:

$$S_{CG} = S_{T_s,V_f^*}, \quad S_{MG} = S_{T_s,V_p^*} + S_{T_w,V_f^*}, \quad S_{FG} = S_{T_w,V_p^*}, \tag{13}$$

and the corresponding similarity losses are denoted as $\mathcal{L}_{CG}$, $\mathcal{L}_{MG}$, and $\mathcal{L}_{FG}$, respectively. By combining these losses, we obtain the total training loss:

$$\mathcal{L}_{total} = \mathcal{L}_{CG} + \alpha\mathcal{L}_{MG} + \beta\mathcal{L}_{FG}, \tag{14}$$

where α and β are the trade-off hyperparameters.

3.4.2. Inference Details

For N query texts and N candidate videos in the test set, we compute the similarity between each query text and all candidate videos, resulting in a $1 \times N$ vector for each query. Repeating this process for all N queries constructs the complete similarity matrix $S \in \mathbb{R}^{N \times N}$. Note that this corresponds to the text-to-video retrieval direction, and by transposing S, we can obtain the video-to-text retrieval direction.

4. Experimental

4.1. Experimental Setup

4.1.1. Datasets

We adopt six benchmark datasets for the evaluation, including MSRVTT [15], DiDeMo [18], LSMDC [19], ActivityNet [27], Charades [28] and VATEX [29]. Table 1 provides the partitioning strategy for these retrieval datasets, and the symbol "-" denotes the absence of a dataset partition. (1) MSRVTT[1] [15] consists of 10,000 YouTube videos, each paired with 20 textual descriptions. The videos in this dataset have medium durations, typically ranging from 10 to 20 seconds. We use the MSRVTT-test-1k split [30], utilizing 9,000 videos for training and 1,000 videos for testing. (2) DiDeMo[2] contains 10,464 video clips and 40,543 captions. We concatenate the descriptions of individual video segments to construct a "video-paragraph" for retrieval. We use the training and testing protocols from MMT [31]. (3) LSMDC[3] [19] contains

[1]http://ms-multimedia-challenge.com/2017/dataset

[2]https://github.com/LisaAnne/LocalizingMoments

[3]https://sites.google.com/site/describingmovies

118,081 video clips from 202 movies. The duration of videos in the LSMDC dataset is short. We follow the split from [19], using 1,000 videos for testing. (4) ActivityNet[4] contains densely annotated temporal segments for 20,000 YouTube videos. Following [14], we report results on the "val_1" split, using 10,009 videos for training and 4,917 for testing. (5) Charades[5] [28] contains 9,848 video clips, totaling around 40 hours of content, with an average duration of 30 seconds per video. Each video is paired with a corresponding text description. We follow the same split protocol as in previous work [9]. (6) VATEX[6] [29] is a large-scale multilingual video description dataset containing over 41,250 videos and 825,000 captions. We follow the training and testing protocol from previous work [9].

Table 1: **Datasets**. Details of the partitioning of six video retrieval benchmarks.

Datasets	Training set		Validation set		Testing set	
	#Videos	#Texts	#Videos	#Texts	#Videos	#Texts
MSRVTT [15]	9,000	180,000	-	-	1,000	1,000
DiDeMo [18]	8,392	8,392	1,065	1,065	1,003	1,003
LSMDC [19]	118,081	118,081	-	-	1,000	1,000
ActivityNet [27]	10,009	10,009	4,926	4,926	4,917	4,917
Charades [28]	7,968	7,968	-	-	1,880	1,880
VATEX [29]	25,991	259,910	1,500	15,000	1,500	15,000

4.1.2. Performance Metrics

Text-video retrieval is commonly evaluated using Recall at rank K (R@K), Median Rank (MdR), and Mean Rank (MnR). R@K, typically measured at

[4]http://activity-net.org/download.html

[5]https://paperswithcode.com/dataset/charades

[6]https://eric-xw.github.io/vatex-website/download.html

1, 5, and 10, indicates the percentage of queries whose correct video appears within the top k results, with higher values reflecting better performance. Additionally, we define R@Sum = R@1 + R@5 + R@10. MdR and MnR represent the median and mean rank of the correct video across all queries, respectively, where lower values denote more accurate retrieval.

4.1.3. Implementation Details

Following previous methods [7, 12], we use CLIP [1] as the backbone model for both text and video feature extraction. The V-Sparse model is trained for 5 epochs with a batch size of 32, and the feature dimension D is set to 512. For the MSRVTT [15], LSMDC [19], and Charades [28] datasets, we uniformly sample $N_f = 12$ frames from each video clip, resize them to $H \times W = 224 \times 224$ pixels, and set the text length to $N_w = 32$. For the DiDeMo [18], ActivityNet [27], and VATEX [29] datasets, we set both $N_f = 64$ and $N_w = 64$. V-Sparse finetunes the CLIP model with a learning rate of 1×10^{-6}, while the remaining modules are trained with a learning rate of 1×10^{-5}. The selected CLIP video encoders include ViT-B/32 and ViT-B/16, with their corresponding image patch sizes P set to 32 and 16, respectively. Training is performed using the AdamW optimizer with a cosine learning rate schedule and a warm-up rate of 0.1. We set the hyperparameters $\tau = 0.01$, $\alpha = 0.5$ and $\beta = 0.1$. For TVSC, the frame selection ratio is set to $\rho_{N_f} = 0.7$, and for SVSC, the patch aggregation ratio is set to $\rho_{N_p} = 0.5$. Experiments are performed on 4 NVIDIA A800 GPUs, with single-GPU training taking approximately 10 hours on MSRVTT and 4.5 hours on DiDeMo. For more details, see the code repository.

4.2. Comparison with State-of-the-art

4.2.1. Comparison Methods

To better validate the effectiveness of V-Sparse, we categorize state-of-the-art methods into three types for comparison with V-Sparse. The first category includes the video retrieval baseline Clip4clip [7]. The second category covers feature enhancement methods, including TS2Net [6], EMCL [32], X-Pool [12], Cap4Video [8], UATVR [33], and T-Mass [9]. The third category consists of feature interaction methods, including DiCoSA [25], CLIP-ViT [24], MSIA [11], HBI [14], DiffusionRet [34], and UCoFiA [13].

4.2.2. Results on the MSRVTT Dataset

In Table 2, we report the performance of the proposed V-Sparse model on both text-to-video and video-to-text retrieval tasks. V-Sparse achieves consistent improvements across two feature extraction backbones, CLIP-ViT-B/32 and CLIP-ViT-B/16. For text-to-video retrieval with the CLIP-ViT-B/32 model, V-Sparse improves R@1 from 50.2 to 50.9 compared to the text-enhancement method T-Mass [9], achieving a gain of **1.4%**. Similarly, with the CLIP-ViT-B/16 model, our R@1 increases from 52.7 to 53.6, resulting in a gain of **1.7%**. These improvements demonstrate that visual compression representation is more effective for cross-modal interaction than textual expansion. In addition, in Table 2, both Cap4Video [8] and CLIP-ViP [24] are text-expansion methods, whose performances in both retrieval directions are consistently lower than that of V-Sparse.

Table 2: Text-to-video and video-to-text retrieval performance on the MSRVTT [15].

Methods	MSRVTT (Text-to-Video)					MSRVTT (Video-to-Text)				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
CE [35]	20.9	48.8	62.4	6.0	28.2	20.6	50.3	64.0	5.3	25.1
MMT [31]	26.6	57.1	69.6	4.0	24.0	27.0	57.5	69.7	3.7	21.3
CLIP-ViT-B/32										
Clip4clip [7]	44.5	71.4	81.6	2.0	15.3	42.7	70.9	80.6	2.0	11.6
EMCL [32]	46.8	73.1	83.1	2.0	-	46.5	73.5	83.5	2.0	-
X-Pool [12]	46.9	72.8	82.2	2.0	14.3	44.4	73.3	84.0	2.0	9.0
DiCoSA [25]	47.5	74.7	83.8	2.0	13.2	46.7	75.2	84.3	2.0	8.9
MSIA [11]	47.2	73.8	84.1	2.0	-	-	-	-	-	-
UATVR [33]	47.5	73.9	83.5	2.0	12.3	46.9	73.8	83.8	2.0	8.6
HBI [14]	48.6	74.6	83.4	2.0	12.0	46.8	74.3	84.3	2.0	8.9
Cap4Video [8]	49.3	74.3	83.8	2.0	12.0	47.1	73.7	84.3	2.0	8.7
CLIP-ViP [24]	50.1	74.8	84.6	1.0	-	-	-	-	-	-
T-Mass [9]	50.2	75.3	85.1	1.0	11.9	47.7	**78.0**	86.3	2.0	8.0
V-Sparse	**50.9**	**76.2**	**85.3**	**1.0**	**11.7**	**49.9**	77.7	**86.4**	**1.0**	**7.1**
CLIP-ViT-B/16										
X-Pool [12]	48.2	73.7	82.6	2.0	12.7	46.4	73.9	84.1	2.0	8.4
UATVR [33]	50.8	76.3	85.5	1.0	12.4	48.1	76.3	85.4	2.0	8.0
Cap4Video [8]	51.4	75.7	83.9	1.0	12.4	49.0	75.2	85.0	2.0	8.0
T-Mass [9]	52.7	77.1	85.6	1.0	10.5	50.9	**80.2**	**88.0**	1.0	7.4
V-Sparse	**53.6**	**79.1**	**87.3**	**1.0**	**9.2**	**52.1**	79.1	87.3	**1.0**	**6.3**

4.2.3. Results on the Long-Video DiDeMo and ActivityNet Datasets

In Table 3, we present the text-to-video retrieval results of V-Sparse on the long-video DiDeMo [18] and ActivityNet [27] datasets. The long-video setting refers to videos longer than 180 seconds, with textual descriptions containing more than 64 words, which serve as important benchmarks for evaluating retrieval performance. For instance, on the DiDeMo dataset, V-Sparse achieves an R@1 of 51.2, outperforming the text-expansion method T-

Table 3: Text-to-video retrieval performance on the long-video datasets DiDeMo [18] and LSMDC [19], as well as the short-text query datasets ActivityNet [27] and Charades [28].

Methods	DiDeMo (Text-to-video)					LSMDC (Text-to-video)				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
EMCL [32]	45.3	74.2	82.3	2.0	12.3	23.9	46.4	53.7	8.0	-
TS2-Net [6]	41.8	71.6	82.0	2.0	14.8	23.4	42.3	50.9	9.0	56.9
X-Pool [12]	44.6	73.2	82.0	2.0	15.4	25.2	43.7	53.5	8.0	53.2
CLIP-ViP [24]	48.6	77.1	84.4	2.0	-	25.6	45.3	54.4	8.0	-
DiCoSA [25]	45.7	74.6	83.5	2.0	11.7	25.4	43.6	54.0	8.0	41.9
DiffusionRet [34]	46.7	74.7	82.7	2.0	14.3	24.4	43.1	54.3	8.0	40.7
T-Mass [9]	50.9	77.2	85.3	1.0	12.1	**28.9**	48.2	57.6	6.0	43.3
V-Sparse	**51.2**	**78.1**	**86.1**	**1.0**	**11.6**	27.4	**48.3**	**58.2**	**6.0**	**39.1**

Methods	ActivityNet (Text-to-video)					Charades (Text-to-video)				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
ClipBERT [5]	21.3	49.0	63.5	6.0	-	6.7	17.3	25.2	32.0	149.7
Clip4clip [7]	40.5	72.4	83.6	2.0	7.5	9.9	27.1	36.8	21.0	85.4
HBI [14]	42.2	73.0	84.6	2.0	6.6	-	-	-	-	-
T-Mass [9]	-	-	-	-	-	14.2	36.2	48.3	12.0	54.8
V-Sparse	**45.1**	**74.5**	**85.0**	**2.0**	**7.3**	**18.2**	**37.5**	**50.4**	**11.0**	**52.3**

Mass [9] by **0.6%**. Furthermore, on the ActivityNet dataset, which contains a large-scale candidate set of 5K videos, V-Sparse surpasses Clip4clip [7] with a remarkable improvement of **11.4%** in R@1. These results clearly demonstrate that feature compression methods are particularly suitable for long-video retrieval and can achieve superior retrieval performance.

4.2.4. Results on the Short-Text LSMDC and Charades Datasets

In Table 4, we present the text-to-video retrieval results of V-Sparse on the short-text LSMDC [19] and Charades [28] datasets. Since the number of words in a single query is less than 10, with only about 5 key target words,

Table 4: Text-to-video retrieval performance on the VATEX [29] dataset.

Methods	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
Support-Set [36]	44.9	82.1	89.7	1.0	-
X-Pool [12]	60.0	90.0	95.0	1.0	3.8
UATVR [33]	61.3	91.0	95.6	1.0	3.3
T-Mass [9]	63.0	92.3	96.4	1.0	3.2
V-Sparse	**80.1**	**97.3**	**98.9**	**1.0**	**1.6**

this poses significant challenges for precise fine-grained matching. On the LSMDC dataset, although V-Sparse performs slightly worse than T-Mass in terms of R@1, it still achieves improvements in overall performance. On the Charades dataset, V-Sparse consistently outperforms all compared methods.

4.2.5. Results on the VATEX Dataset

In Table 5, we present the text-to-video retrieval results of V-Sparse on the VATEX [29] dataset. V-Sparse demonstrates a clear advantage on this dataset, primarily due to each video having 10 rich textual descriptions and a moderate length of around 10 seconds, which allows V-Sparse to effectively leverage video feature compression and fine-to-coarse alignment.

4.3. Ablation Studies

4.3.1. Ablation Study of VSC Module

In Table 5, we divide the visual semantic compression module into three sub-modules, including Temporal Visual Semantic Compression (TVSC), Spatial Visual Semantic Compression (SVSC), and text support (TEXT), to show how different combinations affect the model performance. R0 indicates that no VSC components are used. Adding TVSC in R1 and subsequently TEXT in R2 yields improvements of **1.3** and **2.2** over R0, re-

Table 5: Ablation study of the visual semantic compression (VSC) module on the MSRVTT dataset. "TVSC" and "SVSC" denote the temporal frame and spatial patch feature compression modules, respectively. "TEXT" denotes the text-supported condition.

Rx	TVSC	SVSC	TEXT	MSRVTT (Text-to-Video)				
				R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
R0				45.7	71.8	82.1	2.0	14.6
R1	✓			47.0	72.9	82.2	2.0	14.1
R2	✓		✓	47.9	74.2	83.3	2.0	13.3
R3		✓		49.2	75.6	84.4	2.0	12.8
R4		✓	✓	50.2	74.6	84.6	1.0	12.2
R5	✓	✓	✓	**50.9**	76.2	85.3	1.0	11.7

spectively, demonstrating the effectiveness of global frame-level compression. Adding SVSC in R3 and subsequently TEXT in R4 yields improvements of **3.5** and **4.5** over R0, respectively, demonstrating the effectiveness in local patch-level aggregation. Combining all three sub-modules in R5 achieves a **5.2** improvement over R0, indicating that the proposed components provide strong feature support for subsequent fine-to-coarse interaction.

Note that in the absence of text, R1 employs a frame-similarity aggregation strategy, while R3 uses a self-attention strategy on V_p^* by replacing Q in Equation 7. By comparing R1 with R2 and R3 with R4, we observe performance improvements of **0.9** and **1.0**, respectively, under text query guidance, validating the effectiveness of text-guided visual semantic compression.

4.3.2. Ablation Study of CFI Module

In Table 6, we present the contributions of different coarse-to-fine interaction components to the model performance under visual semantic compression, including coarse, medium, and fine-grained levels. R0 represents the

Table 6: Ablation study of the coarse-to-fine interaction (CFI) modules under the video semantic compression setting. "CG", "MG", and "FG" denote the coarse-grained, medium-grained, and fine-grained interactions in CFI modules, respectively.

Rx	VSC	CFI			MSRVTT (Text-to-Video)				
		CG	MG	FG	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
R0					43.5	68.7	79.6	3.0	16.3
R1		✓			44.5	71.4	81.6	2.0	15.3
R2			✓		46.0	72.9	82.2	2.0	14.8
R3				✓	47.2	71.5	82.4	2.0	14.9
R4	✓	✓			45.7	71.8	82.1	2.0	14.6
R5	✓		✓		47.3	74.3	83.4	2.0	13.8
R6	✓			✓	**48.6**	73.5	84.2	2.0	12.9

baseline retrieval results of Clip4clip [7] using frame-feature average pooling on Equation 1. R1, R2, and R3 show the performance of the three levels of coarse-to-fine interaction, with overall improvements of **1.0**, **2.5**, and **3.7**, respectively. R4, R5, and R6 report the performance of the same interaction levels after applying visual semantic compression, achieving overall improvements of **2.2**, **3.8**, and **5.1**, respectively, further highlighting the strong support provided by VSC for feature interaction.

Table 7: Ablation study of different coarse-to-fine interaction combinations.

Rx	CG	MG	FG	MSRVTT (Text-to-Video)				
				R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
R1	✓	✓		47.2	71.5	82.4	2.0	14.8
R2	✓		✓	48.9	74.5	85.1	2.0	13.3
R3		✓	✓	49.2	75.6	84.4	1.0	12.2
R4	✓	✓	✓	**50.9**	76.2	85.3	1.0	11.7

In Table 7, we report the contributions of different coarse-to-fine interaction combinations to the model performance. Since there are three different interaction levels (coarse, medium, and fine-grained: CG, MG, FG), we explore their performance using pairwise combinations. By comparing different combination strategies, we find that a single combination (e.g., R1) performs worse than the multi-level combination (e.g., R4), which exhibits the strongest complementarity and achieves the highest R@1 of **50.9**.

4.3.3. Ablation Study of the Hyperparameters ρ_{N_f} and ρ_{N_p}

Table 8: Ablation of frame-level feature selection and patch feature aggregation hyperparameters. $N_f = 12$ and $N_p = 49$ represent the initial number of video frames and image patches, respectively, while N_f^* and N_p^* denote the number of features after compression.

Rx	ρ_{N_f}	$N_f \rightarrow N_f^*$	ρ_{N_p}	$N_p \rightarrow N_p^* \rightarrow N_p^* \rightarrow N_p^*$	MSRVTT (Text-to-Video)				
					R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
R1	100%	$12 \rightarrow 12$	100%	$49 \rightarrow 49 \rightarrow 49 \rightarrow 49$	45.7	71.8	82.1	2.0	14.6
R2	90%	$12 \rightarrow 10$	90%	$49 \rightarrow 45 \rightarrow 41 \rightarrow 37$	47.5	73.7	83.0	2.0	12.0
R3	80%	$12 \rightarrow 09$	70%	$49 \rightarrow 35 \rightarrow 25 \rightarrow 18$	47.5	74.0	83.3	2.0	11.8
R4	70%	$12 \rightarrow 08$	50%	$49 \rightarrow 25 \rightarrow 13 \rightarrow 07$	**50.2**	77.6	85.4	1.0	11.2
R5	60%	$12 \rightarrow 07$	30%	$49 \rightarrow 15 \rightarrow 05 \rightarrow 02$	49.7	76.0	84.8	1.0	11.6
R6	50%	$12 \rightarrow 06$	10%	$49 \rightarrow 05 \rightarrow 01 \rightarrow 01$	48.3	75.3	84.4	2.0	12.2

ρ_{N_f} and ρ_{N_p} denote the proportions after frame feature selection and patch feature aggregation, respectively. In Table 8, we set the number of retained frames as $N_f^* = \rho_{N_f} \times N_f$, and the number of patch clusters as $N_p^* = \rho_{N_p} \times N_p$ to evaluate the impact of different parameter settings on retrieval performance, with initial values $N_f = 12$ and $N_p = 49$. We observe that the parameter setting in R4 achieves the best performance with R@1 = 50.2%, while the other settings yield inferior results due to being either too high or too low. Compared with R4, although R6 retains 50% of the frames, only 10% of patches are kept per frame, resulting in a significantly reduced number of features involved in granular interactions ($N_f^* = 6$, $N_p^* = 6$). Excessive compression (R6) can lead to the loss of critical visual information and, consequently, a substantial drop in performance, whereas a moderate level of feature compression (R4) effectively removes redundancy while preserving key semantic information, thereby improving retrieval performance.

4.3.4. Ablation Study of Hyperparameter N_f

N_f denotes the average number of frames sampled per video, which measures the temporal sampling density of the video. In Figure 5, we set different frame numbers $N_f = \{12, 15, 18, 21, 24\}$ and compare the performance of V-Sparse with two state-of-the-art methods, X-Pool [12] and T-Mass [9], on the Charades [28] to evaluate the performance of V-Sparse under long video sequence sampling. As the number of sampled frames increases, the model's performance improves, but so does the computational cost. Therefore, to achieve better retrieval results, N_f can be appropriately increased, especially when processing long videos such as DiDeMo [18] and ActivityNet [27].

4.4. Further Analysis

4.4.1. Analysis of Similarity under Compression

In Sec. 3.2, we analyze the similarity between visual features before and after compression, demonstrating the effectiveness of the compression for both visual frame and patch representations. To further illustrate this, we compute the similarity between the text and video features before and after compression. The first case considers the similarity for related text-video

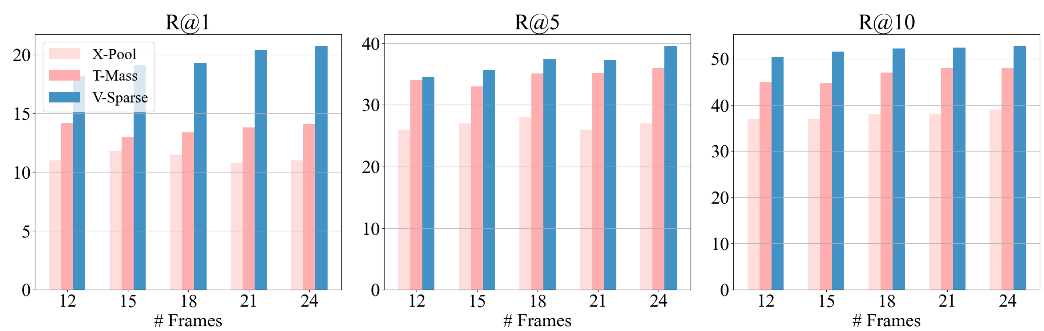


Figure 5: Ablation of different sampling frame numbers on the Charades [28].

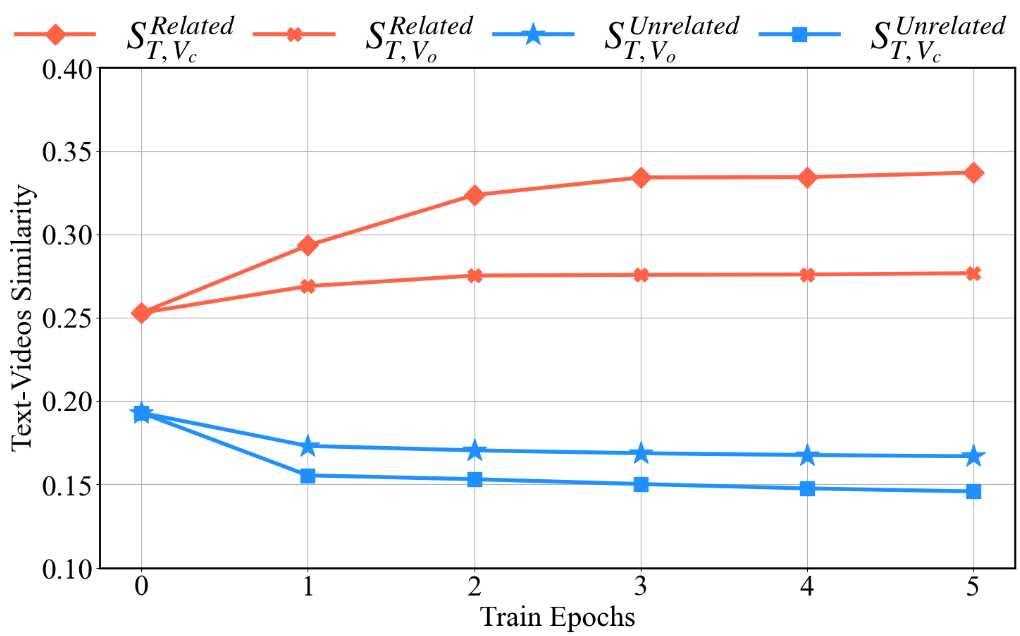


Figure 6: **Text-video similarity.** The similarities between the related and unrelated text-video pairs are denoted by $S_{T,V}^{Related}$ and $S_{T,V}^{Unrelated}$, respectively. A larger $S_{T,V}^{Related}$ is better, while a smaller $S_{T,V}^{Unrelated}$ is better. The representations of V include both the original visual feature V_o and the compressed visual feature V_c.

pairs, including $S_{T,V_o}^{Related}$ and $S_{T,V_c}^{Related}$. We compute the average similarity for all 1,000 related text-video pairs in the MSRVTT dataset, as shown by the top two curves in Figure 6. The second case considers the similarity for unrelated text-video pairs, including $S_{T,V_o}^{Unrelated}$ and $S_{T,V_c}^{Unrelated}$. We compute the average similarity for all unrelated text-video pairs, as shown by the bottom two curves in Figure 6. Upon observation, we find that using V_c increases the similarity between the video and related text while decreasing the similarity with unrelated text. This further demonstrates the effectiveness of video semantic compression in improving alignment with text.

4.4.2. Analysis of Cross-Domain and Cross-Task Performance

Table 9: Analysis of cross-domain generalization performance. We use $X > Y$ to denote training on dataset X and testing on dataset Y, where X is always MSRVTT, and Y corresponds to MSRVTT, DiDeMo, and LSMDC, respectively. R@Sum=R@1+R@5+R@10.

Methods	MSRVTT>MSRVTT			MSRVTT>DiDeMo			MSRVTT>LSMDC		
	R@1↑	R@Sum↑	MdR↓	R@1↑	R@Sum↑	MdR↓	R@1↑	R@Sum↑	MdR↓
Clip4clip [7]	44.5	197.5	2.0	31.8	154.9	4.0	15.3	87.1	21.0
X-Pool [12]	46.9	201.9	2.0	35.3	168.5	3.0	16.4	93.5	18.0
EMCL [32]	47.0	203.0	2.0	30.0	151.9	4.0	16.6	82.1	24.0
DiffusionRet [34]	49.0	206.9	2.0	33.2	160.9	3.0	17.1	90.5	21.0
T-Mass [9]	50.2	210.6	1.0	39.5	178.2	2.0	19.7	102.5	14.0
V-Sparse	**50.9**	212.4	1.0	**42.0**	185.6	2.0	**20.4**	108.2	12.0

To further evaluate the model's cross-domain generalization capability on unseen datasets, we present the results of out-of-domain retrieval in Table 9. Specifically, the V-Sparse and state-of-the-art models are first pre-trained on the "source" dataset X, e.g., MSRVTT [15], and then evaluated on the "target" dataset Y, such as the long-video dataset DiDeMo [18] and the short-text dataset LSMDC [19]. We find that models performing well on in-domain training data experience significant performance drops when generalized to out-of-domain datasets. However, compared with these models, V-Sparse achieves the best performance (R@1: 42.0% and 20.4%) on both the long-video dataset DiDeMo and the short-text dataset LSMDC, demonstrating its strong generalization capability and practical effectiveness in representing long video sequences and short-text queries.

Table 10: Analysis of cross-task generalization performance in video question answering.

Methods	Publications	Accuracy(%)↑
VQA-T [37]	ICCV2021	41.5
EMCL-QA [32]	NIPS2021	45.8
MERLOT [38]	CVPR2022	43.1
Co-Tokenization [39]	ECCV2022	45.7
HBI [14]	CVPR2023	46.2
V-Sparse	-	**47.5**

To further demonstrate the generality of the visual semantic compression method, we extend it to the cross-domain video question answering task, as shown in Table 10. Compared with the hierarchical visual-question-answer alignment method HBI [14], V-Sparse achieves a 2.8% improvement in accuracy, demonstrating strong alignment with key semantic content and significant benefits from textual guidance.

4.4.3. Analysis of Visual Semantic Noise

In Table 11, we show the impact of different Gaussian noise standard deviations σ on visual feature representations, i.e., $f = f + \sigma$, where larger values of σ can disrupt feature semantics and negatively affect performance. Compared with the visual feature enhancement method X-Pool [12] and the textual feature enhancement method T-Mass [9], V-Sparse maintains stable

Table 11: Ablation study of different Gaussian noise standard deviations σ.

Noise	Methods	MSRVTT (Text-to-Video)				
		R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
-	X-Pool [12]	46.9	72.8	82.2	2.0	14.3
-	T-Mass [9]	50.2	75.3	85.1	1.0	11.9
-	V-Sparse (Ours)	**50.9**	76.2	85.3	1.0	11.7
$\sigma = 0.01$	X-Pool [12]	43.7	68.5	79.5	3.0	15.7
$\sigma = 0.01$	T-Mass [9]	48.7	73.4	82.1	2.0	12.5
$\sigma = 0.01$	V-Sparse (Ours)	49.1	74.6	83.5	2.0	12.5
$\sigma = 0.1$	X-Pool [12]	40.3	64.5	74.2	4.0	16.8
$\sigma = 0.1$	T-Mass [9]	46.7	70.5	76.5	3.0	14.5
$\sigma = 0.1$	V-Sparse (Ours)	47.8	70.8	78.5	3.0	13.9
$\sigma = 0.3$	X-Pool [12]	36.8	60.4	72.1	6.0	19.6
$\sigma = 0.3$	T-Mass [9]	42.3	65.7	73.1	5.0	16.4
$\sigma = 0.3$	V-Sparse (Ours)	45.1	67.8	74.6	4.0	15.4

performance across a range of noise settings $\sigma \in \{0.01, 0.1, 0.3\}$. For example, under a high-noise setting ($\sigma = 0.3$), the R@1 performance of X-Pool and T-Mass drops by **10.1%** and **7.9%**, respectively, whereas V-Sparse decreases by only **5.8%**, indicating its strong robustness to feature noise.

4.4.4. Analysis of Retrieval Efficiency

In Table 12, considering that V-Sparse integrates three granularity alignment strategies, we compare the retrieval efficiency of our method with X-Pool [12] and T-Mass [9]. Specifically, we compare aspects such as model sampling time, parameters, and performance metrics, while evaluating V-Sparse with three different coarse-to-fine alignment combinations. Compared to the text-expansion method T-Mass, V-Sparse demonstrates a significant advantage in retrieval speed and achieves superior retrieval metrics.

Table 12: Analysis of retrieval efficiency. We compare the differences between the components of V-Sparse and state-of-the-art models in terms of sampling time, model parameters, and retrieval performance. All model sampling is performed on an NVIDIA RTX 3090.

Methods	Sample Time↓	Parameters↓	MSRVTT (Text-to-Video)				
			R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
X-Pool [12]	157.143s	∼152.6M	46.9	73.2	82.0	2.0	15.4
T-Mass [9]	271.247s	∼153.4M	50.2	75.3	85.1	1.0	11.9
CG + MG	200.345s	∼153.4M	47.2	71.5	82.4	2.0	14.8
CG + FG	210.134s	∼155.7M	48.9	74.5	85.1	2.0	13.3
MG + GF	215.145s	∼156.2M	49.2	75.6	84.4	1.0	12.2
V-Sparse	225.763s	∼157.9M	**50.9**	76.2	85.3	1.0	11.7

4.5. Visualization

4.5.1. Visualization of Compression Process

In Figure 7, we visualize the video frame selection process. Frames in the red-line regions, which have low similarity with the text query, are removed along with their corresponding patch-level visual features. The bottom S represents the similarity S_{T_s,V_f^i} defined in Equation 1.

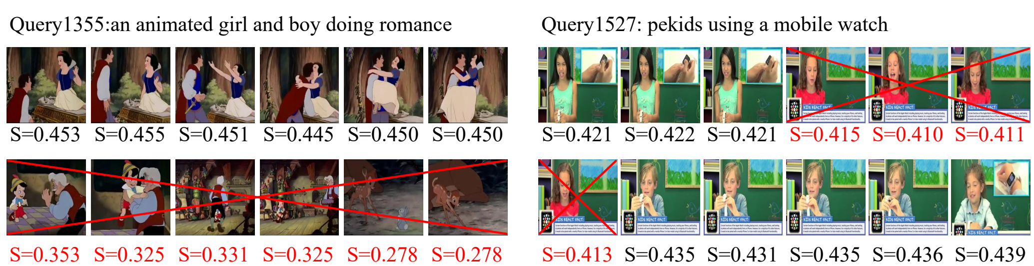


Figure 7: Visualization of the video frame selection process. Given a query text and sampled video frames, the frames with the highest similarity scores are retained.

In Figure 8, we visualize the frame patch compression process. Vanilla tokens and ρ_{N_p} denote the original, uncompressed visual patches and the ratio of compressed to original visual patches, respectively. By setting different values of ρ_{N_p}, a large number of patch features are continuously merged, not only significantly reducing the number of video semantic features, but also preserving complete visual entity semantic characteristics.

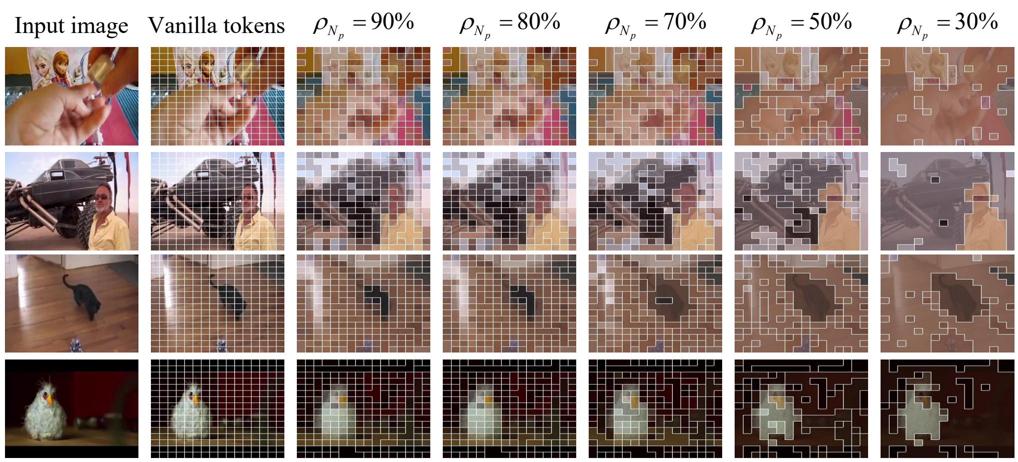


Figure 8: Visualization of the frame patch compression process. ρ_{N_p} denotes the ratio of the number of features after compression to that before compression.

In Figure 9, we visualize the frame patch compression process with text support. We can intuitively observe that, under the guidance of textual semantics, visual features focus more on key entities rather than being aggregated in a dispersed manner, demonstrating that the compression process achieves stronger feature aggregation when guided by text.

4.5.2. Visualization of Feature Interaction Process

In Figure 10, we present the quantitative alignment results of weights and similarity values. We observe that medium-grained alignment dynamically adjusts coarse-grained frame weights, focusing on more informative details, such as emphasizing the *"robot"* over the *"person"*.

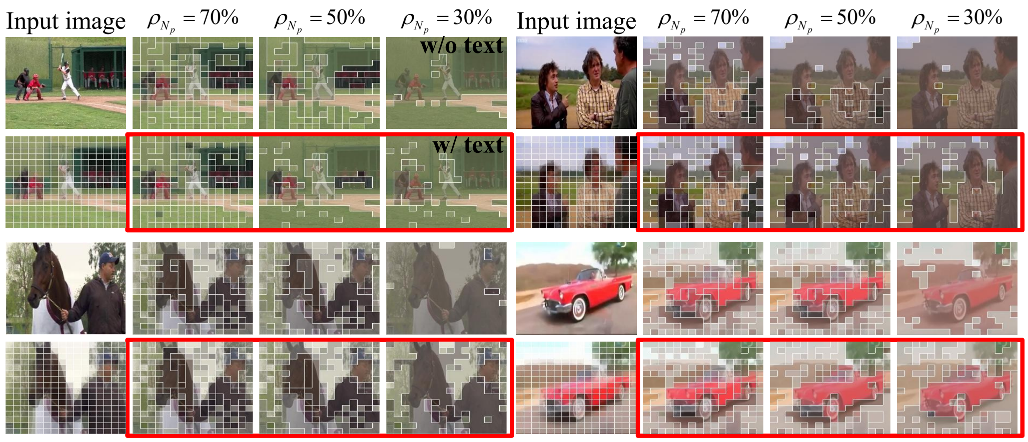


Figure 9: Visualization of patch compression with text support. The red box (w/ text) indicates the results obtained with textual guidance.

4.5.3. Visualization of Retrieval Result

In Figures 11 and 12, we visualize specific query cases and their retrieval results benefiting from video semantic compression. Specifically, in Figure 11, given the query "A little girl does gymnastics," we show the retrieval rankings of X-Pool and V-Sparse in (a) and (b), respectively. In Figure 12, given the query "A man is singing on the road," we show the retrieval rankings of T-Mass and V-Sparse in (a) and (b), respectively. Benefiting from visual compression and granular alignment, only V-Sparse retrieves the correct result, achieving R@1 = **100%**, while all other methods achieve **0%**.

In Figure 13, we further visualize a set of failed retrieval results. The left side shows the videos matched to the given queries in the MSRVTT dataset, while the right side shows the results retrieved by our method, V-Sparse. It is evident that these query sentences are very sparse and do not provide a sufficient description of the videos, which is the primary cause of retrieval

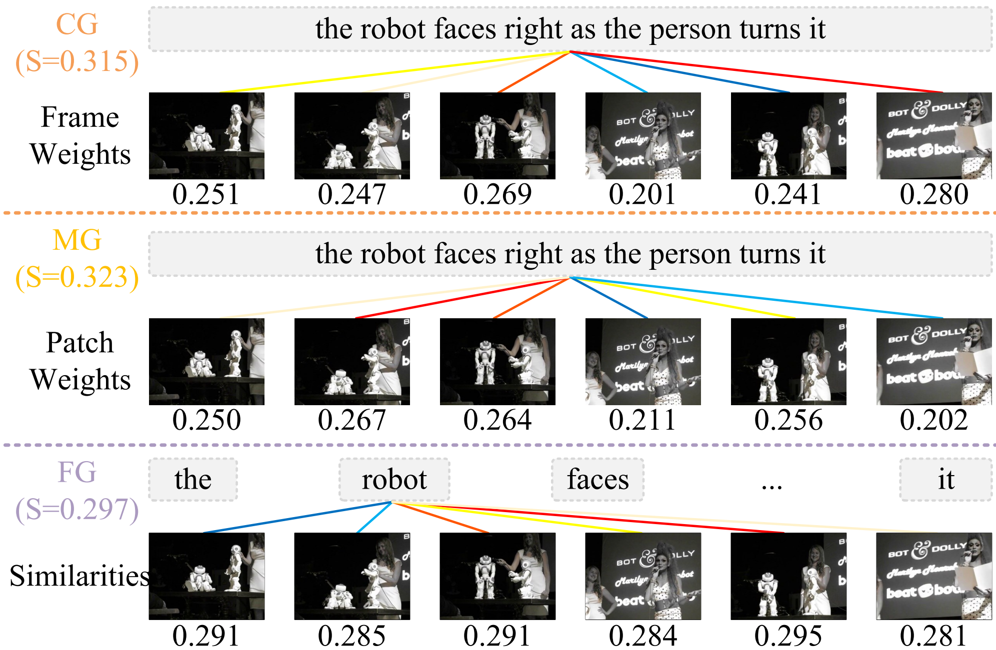
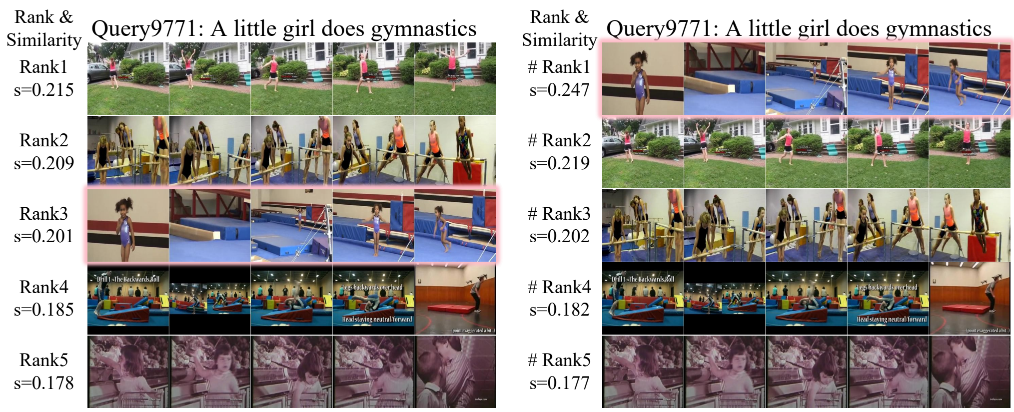


Figure 10: Visualization of the weight and similarity values in the coarse-to-fine interaction. The weight and similarity are represented in the following colors from high to low: red, light, red, yellow, light yellow, blue, and light blue, respectively.

failures. Moreover, our retrieved results partially match the textual queries; for example, for "Query7581: A man prepares some food in the kitchen," the entity "A man" is correctly matched. These failures are attributed to the inherent limitations of the dataset rather than to the model itself. In the future, we will further investigate this issue to improve retrieval performance.

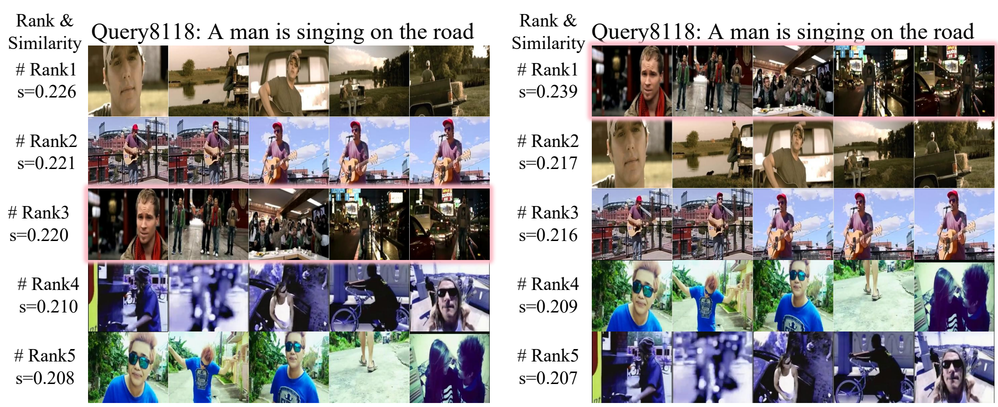
5. Conclusion

In this paper, to address the challenge of interacting rich visual information with sparse text, we propose a novel text-video retrieval model named V-Sparse, which incorporates visual semantic compression for feature en-



(a) Retrieval results ranking of **X-Pool** (b) Retrieval results ranking of **V-Sparse**

Figure 11: Visualization of X-Pool and V-Sparse retrieval results. The ground-truth result is highlighted in pink, and the ranking is based on similarity scores.



(a) Retrieval results ranking of **T-Mass** (b) Retrieval results ranking of **V-Sparse**

Figure 12: Visualization of T-Mass and V-Sparse retrieval results. The ground-truth result is highlighted in pink, and the ranking is based on similarity scores.

hancement and a coarse-to-fine alignment strategy for feature interaction. By incorporating a text-supported visual semantic compression module, we dynamically compress visual features across both the temporal frames-level and the spatial patches-level, thereby enhancing their expressiveness and flexibil-

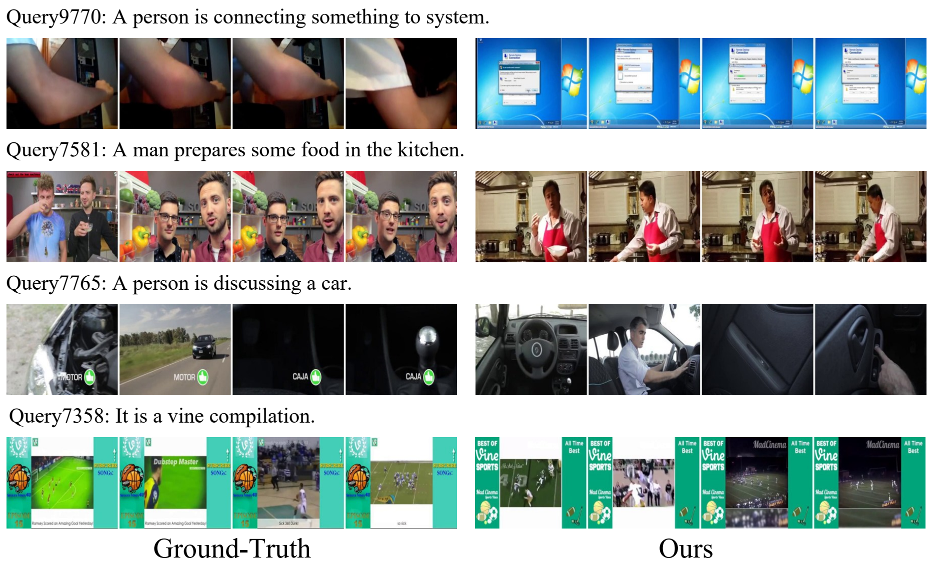


Figure 13: Visualization of failed retrieval cases by V-Sparse. The left side shows the ground-truth matches, while the right side shows the results retrieved by our method.

ity for text matching. Benefiting from the compressed visual semantics, we further introduce a novel coarse-to-fine granularity interaction mechanism, which progressively aligns features at the sentence–frame, sentence–patch, and word–patch levels, ensuring comprehensive and robust cross-modal feature correspondence. We evaluate the retrieval performance of V-Sparse on six benchmark datasets, including MSRVTT, DiDeMo, LSMDC, ActivityNet, Charades, and VATEX, achieving state-of-the-art results. We hope that this work will further advance the field of text-video retrieval.

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